

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	3

Case Study Topic: Indian Railways Bharat Gauge Upgrade using Kruskal's Minimum Spanning Tree (MST) Algorithm

Word Done Summary:

In this case study, Kruskal's Algorithm was applied to determine the minimum-cost railway electrification network connecting six major Indian railway stations.

Implementation Details:

1. Edge sorting by cost.
2. Union-Find (Disjoint Set Union) data structure.
3. Cycle detection.
4. MST construction.
5. Total cost computation.
6. Comparison with Prim's Algorithm.

Result:

Sorted Edge List :

BPL-NGP 350

NGP-HYD 500

HYD-CHN 600

DEL-BPL 700

BPL-MUM 800

MUM-NGP 850

NGP-CHN 1100

DEL-MUM 1400

Accepted Edges :

BPL-NGP

NGP-HYD

HYD-CHN

DEL-BPL

BPL-MUM

Rejected Edges :

MUM-NGP

NGP-CHN

DEL-MUM

Reason :

Cycle Formation

Output:

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Edges in MST:  
BPL - NGP : 350  
NGP - HYD : 500  
HYD - CHN : 600  
DEL - BPL : 700  
BPL - MUM : 800  
  
Total Cost = 2950
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GitHub Link:

<https://github.com/saisharan134/DSA2-Case-Studies>

Student Signature	Faculty Signature
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